

WEEK-5:JDBC CRUD OPERATIONS

Aim

Develop a Java standalone application that connects with the database (Oracle / MySQL) and perform the CRUD operation on the database tables.

Learning Objectives of this experiment

- To understand how Java applications interact with relational databases using JDBC architecture and drivers.
- To learn how to establish a connection between a standalone Java application and MySQL/Oracle database.
- To design and execute Create, Read, Update, and Delete (CRUD) operations using SQL queries in Java.
- To gain knowledge of exception handling and proper management of database resources.
- To understand the integration of backend application logic with database tables for effective data management.

CODE:

```
import java.sql.*;
import java.util.*;

public class StudentCRUD {
    static final String DB_URL = "jdbc:oracle:thin:@10.100.0.11:1521:ORCL11G";
    static final String USER = "24071A0569";
    static final String PASS = "24071A0569";

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Connection conn = null;
        try {
            Class.forName("oracle.jdbc.driver.OracleDriver");
            conn = DriverManager.getConnection(DB_URL, USER, PASS);
            System.out.println("Connected to Oracle Database Successfully!");
            createTable(conn);
            int choice;
            do {
                System.out.println("\n===== STUDENT CRUD MENU =====");
                System.out.println("1. Insert Student");
                System.out.println("2. Display All Students");
```

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System.out.println("3. Update Student Marks");
System.out.println("4. Delete Student");
System.out.println("5. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {
    case 1 -> insertStudent(conn, sc);
    case 2 -> displayStudents(conn);
    case 3 -> updateStudent(conn, sc);
    case 4 -> deleteStudent(conn, sc);
    case 5 -> System.out.println("Exiting...");
    default -> System.out.println("Invalid choice!");
}
} while (choice != 5);

} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
} finally {
    try {
        if (conn != null) conn.close();
        System.out.println("Connection closed.");
    } catch (SQLException e) {
        System.out.println("Error closing connection: " + e.getMessage());
    }
}
}

static void createTable(Connection conn) throws SQLException {
    Statement stmt = conn.createStatement();
    try {
        String sql = "CREATE TABLE student (" +
            "regno NUMBER PRIMARY KEY, " +
            "name VARCHAR2(50), " +
            "mark1 NUMBER, " +
            "mark2 NUMBER, " +
            "mark3 NUMBER)";
        stmt.executeUpdate(sql);
        System.out.println("Table 'student' created successfully.");
    } catch (SQLException e) {
        System.out.println("Table already exists, continuing...");
    }
}
```

```
}
stmt.close();
}

static void insertStudent(Connection conn, Scanner sc) throws SQLException {
    String sql = "INSERT INTO student VALUES (?, ?, ?, ?, ?)";
    PreparedStatement pstmt = conn.prepareStatement(sql);
    System.out.print("Enter Register No: ");
    int reg = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Name: ");
    String name = sc.nextLine();
    System.out.print("Enter Mark1: ");
    int m1 = sc.nextInt();
    System.out.print("Enter Mark2: ");
    int m2 = sc.nextInt();
    System.out.print("Enter Mark3: ");
    int m3 = sc.nextInt();
    pstmt.setInt(1, reg);
    pstmt.setString(2, name);
    pstmt.setInt(3, m1);
    pstmt.setInt(4, m2);
    pstmt.setInt(5, m3);

    int rows = pstmt.executeUpdate();
    System.out.println(rows + " record inserted successfully!");
    pstmt.close();
}

static void displayStudents(Connection conn) throws SQLException {
    Statement stmt = conn.createStatement();
    ResultSet rs = stmt.executeQuery("SELECT * FROM student");
    System.out.println("\n--- Student Details ---");
    while (rs.next()) {
        System.out.println("RegNo: " + rs.getInt("regno") +
            ", Name: " + rs.getString("name") +
            ", Marks: " + rs.getInt("mark1") + ", " +
            rs.getInt("mark2") + ", " + rs.getInt("mark3"));
    }
    rs.close();
    stmt.close();
}
```

```
}

static void updateStudent(Connection conn, Scanner sc) throws SQLException {
    String sql = "UPDATE student SET mark1=?, mark2=?, mark3=? WHERE regno=?";
    PreparedStatement pstmt = conn.prepareStatement(sql);
    System.out.print("Enter Register No to Update: ");
    int reg = sc.nextInt();
    System.out.print("Enter New Mark1: ");
    int m1 = sc.nextInt();
    System.out.print("Enter New Mark2: ");
    int m2 = sc.nextInt();
    System.out.print("Enter New Mark3: ");
    int m3 = sc.nextInt();
    pstmt.setInt(1, m1);
    pstmt.setInt(2, m2);
    pstmt.setInt(3, m3);
    pstmt.setInt(4, reg);
    int rows = pstmt.executeUpdate();
    if (rows > 0)
        System.out.println("Record updated successfully!");
    else
        System.out.println("Student not found!");
    pstmt.close();
}

static void deleteStudent(Connection conn, Scanner sc) throws SQLException {
    String sql = "DELETE FROM student WHERE regno=?";
    PreparedStatement pstmt = conn.prepareStatement(sql);

    System.out.print("Enter Register No to Delete: ");
    int reg = sc.nextInt();

    pstmt.setInt(1, reg);
    int rows = pstmt.executeUpdate();
    if (rows > 0)
        System.out.println("Record deleted successfully!");
    else
        System.out.println("Student not found!");
    pstmt.close();
}
```

}

OUTPUT:

Connected to Oracle Database Successfully!

===== STUDENT CRUD MENU =====

1. Insert Student
2. Display All Students
3. Update Student Marks
4. Delete Student
5. Exit

Enter your choice: 1

Enter Register Number: 22071

Enter Name: Rahul

Enter Mark 1: 85

Enter Mark 2: 90

Enter Mark 3: 78

Student inserted successfully!

Enter Register Number: 22072

Enter Name: Priya

Enter Mark 1: 92

Enter Mark 2: 88

Enter Mark 3: 95

Student inserted successfully!

--- Student Information ---

Reg No: 22071, Name: Rahul, Mark1: 85, Mark2: 90, Mark3: 78

Reg No: 22072, Name: Priya, Mark1: 92, Mark2: 88, Mark3: 95

Enter your choice: 3

Enter Register Number to update: 22071

Enter new Mark1: 88

Enter new Mark2: 92

Enter new Mark3: 81

Student record updated successfully!

Enter your choice: 4

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Name of the Laboratory: _____

Name of the Experiment: _____

Experiment No: _____ Date: _____

Enter Register Number to delete: 22072

Student deleted successfully!

--- Student Information ---

Reg No: 22071, Name: Rahul, Mark1: 88, Mark2: 92, Mark3: 81

Enter your choice: 5

Exiting...

Connection closed.